

A State-of-the-Art Review of the Implementation, Challenges, and Future Opportunities of Quantum Internet Networks

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Abstract—Since its emergence, the field of quantum mechanics has slowly upended the way we perceive and understand our world. As the study of quantum information and communication progresses, anticipation for its possible integration in modern-day communication networks grows, paving the way toward the Quantum Internet. However, quantum technology is difficult to implement in practice, largely due to qubits having complex and unusual properties.

The aim of this thesis is to provide a survey on the ongoing development of quantum technologies, particularly in the context of Internet networks. It examines state-of-the-art research in the field of quantum communications with the intent to summarize and discuss its present and future implementations. Covering the former, this thesis first investigates quantum applications in contemporary telecommunication infrastructure. It also seeks to formalize the challenges that quantum mechanisms introduce to the classical communication network while analyzing potential improvements and solutions. Finally, the thesis explores the use of quantum mechanisms in the development of a dedicated Quantum Internet protocol stack, focusing on the opportunities it can bring to the telecommunication field as well as proposals in its composition and operation.

Keywords—quantum networking, quantum internet, protocol stack

I. INTRODUCTION

With the advancements of classical technology reaching their physical limits—an observation aptly demonstrated with Moore’s law, where the number of transistors on integrated circuits have doubled every two years since the late 20th century [1], [2]—micro-electronic components have reached a size where the effects of subatomic (or quantum) particles can no longer be ignored. As a result, the importance of quantum research and related emerging technologies continues to grow.

One such ongoing field of research is posed to succeed the modern-day Internet and its relevant systems: quantum networking. From the turn of the century, communication networks have evolved across several generations, with 6G being the latest—and potentially final, as [3] hypothesizes—in the line of completely classical mobile communications technology before quantum interplay. Indeed, there have been various achievements in experimental implementations of quantum networks, with researchers worldwide realizing large-scaled networks with the use of existing telecommunication infrastructure, thus validating the feasibility of a quantum internet architecture [4]. Notably, most state-of-the-art implementations are focused on quantum key distribution networks, as the maturity of other components necessary for cloud quantum computing, and

eventually a quantum internet, such as the quantum repeater, are still in their infancy.

This paper aims to provide a review of existing literature, providing a snapshot of ongoing developments pertaining to quantum internet networks. The next section provides brief preliminaries on classic network architecture in place today alongside the necessary quantum fundamentals needed to understand the concepts touched on in this thesis. Then, Section III covers existing infrastructure of classical networks and the feasibility—and eventual complications—of enabling quantum applications in contemporary telecommunication systems. Section IV will aim to explore the use of quantum mechanisms and computational power in the development of a new Quantum Internet protocol stack. Finally, Section V provides concluding remarks.

II. PRELIMINARIES

A. Classical Networking

Digital communication in the present day takes place largely on the Internet, a globally-spanning series of interconnected networks where two or more hosts can exchange information.

The Internet is an example of a packet-switched (also known as connectionless) network. This means that each individual packet of information is sent separately and independently, regardless if it is part of a larger message. Crucially, there is no relationship established between sender and receiver before the start of transmission. This is antithetical to circuit-switched networks (also known as connection-oriented), where sender and receiver must establish a physical connection before initiating communication.

Two abstract models that were used in the design of the Internet are the OSI model and the TCP/IP model [5]. Together, they form the protocol suite, essentially a set of rules, used in the Internet today. Composed of five layers, each upper-layer protocol depends on the services obtained from a lower-layered one [6].

Oversimplifying, to transmit a piece of information from one device to another, messages are “sent down” from the application layer to the physical layer where it is encoded into binary bits, made into a signal, and sent across a transmission medium (cable or free space) towards its destination. Fig. 1 illustrates a generic communication between two devices through an internet, which highlights the usage of two supporting devices: a switch and a router, the latter being essential for long-range communication as it forwards data across large geographical areas towards a particular address in a process called routing.

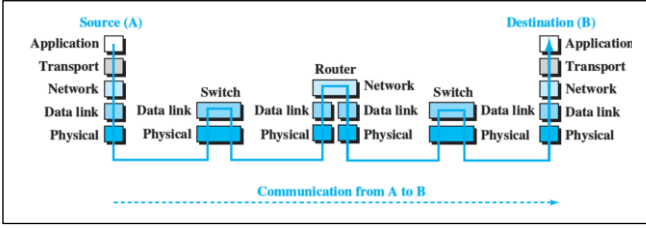


Fig. 1. Model of a generic conversation between two parties across an internet following the joint OSI and TCP/IP protocol suite. Taken and cropped from [6].

Ultimately, classical digital communication, at its core, is deterministic. The fundamental unit of information, the binary bit, can either be a 0 or a 1 which means any potential error in transmission will result in an erroneous bit, called a bit flip. Indeed, communication through any non-ideal, realistic environment will introduce error to the overall system. However, within the Internet’s protocol stack, there are several contingencies scattered across different layers to increase the overall reliability of the system and avoid, or otherwise catch and deal with, errors. These functions are found within the data link layer and the transport layer and implement practices like Automatic ReQuests (ARQ) and flow control.

Additionally, as a global service, the Internet must also manage multiple devices’ simultaneous requests for the communication channel. This dilemma is also addressed within the data link layer in a sublayer called Medium Access Control (MAC) with dedicated protocols split between three forms access: random, controlled, and channelized [6].

B. Quantum Information Fundamentals

The core difference between classical and quantum communications is found in the introduction of the quantum bit, or qubit for short. Qubits, contrary to classical binary bits which are found in either a 0 or 1 state, can be defined in a linear combination of states. The most common notation for expressing qubits is called Dirac, or bra-ket, notation:

$$|\varphi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (1)$$

The general state of a qubit $|\varphi\rangle$ contains the superposition of the both classical, or basis, states which are directly correlated to their respective probability amplitudes $\alpha, \beta \in \mathbb{C}$. Upon measuring a quantum bit, it will collapse into one of the two states with a probability of $|\alpha|^2$ or $|\beta|^2$, respectively.

Furthermore, these complex constants must comply with the normalization condition $|\alpha|^2 + |\beta|^2 = 1$ and can be extended to multiple qubit systems which would eventually form the basis of transmitted data packets. In fact, there is an important distinction to be made between physical and logical qubits. The former is an elementary quantum bit realized in physical form (e.g. photons, superconductive materials, trapped ions), while the latter is an abstraction of a higher level where multiple physical qubits can be grouped together in an effort to reduce errors, leading the way towards error-tolerant quantum schemes [7].

Quantum information is, by nature, fragile. Akin to the presence of noise in the classical domain, decoherence refers to the irreversible erosion of a quantum state typically caused by unwanted interactions with the environment. However,

qubits, when compared to their classical counterpart, are substantially more susceptible to imperfections—as well as any contamination via their handling—due to quantum-domain noise being a multiplicative defect [8]. A strategy implemented in the classical domain, the creation and storage of redundant copies of the same piece of information, is not allowed in the quantum domain. This characteristic is aptly named as the no-cloning theorem.

These disadvantages pose a non-trivial challenge in the design of network systems, as it presents two pressing issues. First, decoherence introduces a timing constraint that limits how long a given quantum task must be completed before quantum resources are no longer usable. Secondly, the no-cloning theorem asserts that quantum information cannot be stored in two locations at the same time.

Before detailing how quantum information is transmitted, we first introduce a core phenomenon in quantum physics with no classical analogy: entanglement. This phenomenon intertwines the state of two or more qubits such that the system is innately dependent on each other, regardless of physical distance. Entanglement, and its extension in teleportation, rely on so-called maximally entangled states in the form of Einstein-Podolsky-Rosen (EPR) pairs or, for generic numbered qubit systems, Greenberger–Horne–Zeilinger (GHZ) state and the W-state [7], [9], [10]. For simplicity, we focus on the EPR pair, modeled by:

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (2)$$

which could also be modeled using $|01\rangle$ and $|10\rangle$, forming the four basis states of a two-qubit system.

EPR pairs, also known Bell states, are widely used in quantum communication due to the fact that they can be transformed into any arbitrary state through local operations. These operations, known as quantum gates, are fundamental tool in the manipulation of quantum systems. Notably, they are linear, commonly reversible, and do not erase any information [9], [11]. As such, as any generic pair of qubits can be expressed as a linear combination of the four Bell basis states.

Before measuring an entangled system, it is impossible to know its quantum state. However, if the first qubit to be measured (the probability for either state modelled by $(\frac{1}{\sqrt{2}})^2$, i.e., a 50% probability for either state) is $|0\rangle$, then its pair, once measured, will produce the same outcome of $|0\rangle$ every single time. This instantaneity makes entanglement a powerful resource and opens the door to new possibilities in communication.

Quantum teleportation is a fundamental and critical piece of the quantum communication puzzle which leverages entanglement properties. Particularly, we focus on the case where Alice wants to send a generic, unknown quantum state $|\varphi\rangle$ to Bob. Teleportation protocol allows for the transfer of a quantum state without any physical displacement of the original particle nor violation of quantum mechanics.

In essence, a quantum link is established between communicating parties through entanglement distribution. Additionally, on Alice’s side, a quantum state is prepared which contains the piece of quantum information intended for transmission. Upon joint measurement of both her information

qubit and one half of the entangled pair (collapsing Bob's entangled bit to state $|\varphi\rangle$), two classical bits are obtained and transmitted to Bob over classical communication channels. Finally, with both classical bits, Bob can perform a series of quantum gates in order to recover the original information quantum state $|\varphi\rangle$.

Importantly, the measurements destroy the original quantum state at Alice, as well as the entangled pair originally distributed. Furthermore, as it is noted in [8], this transmission of information does not exceed the faster-than-light limitations set by the relativity theory since, without the classical transmission from Alice, the original state could not be otherwise obtained.

Finally, in two relevant extensions, we detail how quantum communication can be reinforced in realistic transmissions settings. **Entanglement swapping** is the act of entangling two quantum systems that have previously never interacted with the other, and it is the foundation for realizing a **quantum repeater**. As illustrated in Fig. 2, swapping can be used for cutting distance in the distribution of entangled particles, allowing for locally produced and distributed pairs (in say, a quantum repeater) to be used in order to link nodes across a much larger distance. Additionally, with the expressed aim of obtaining higher quality entanglement states, the **entanglement purification** procedure combines various, lower quality entanglements into a single near-maximally entangled state.

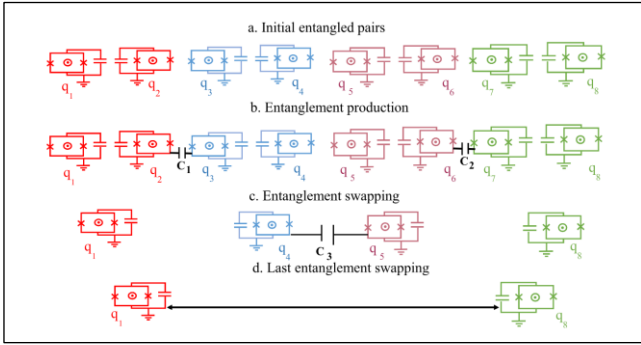


Fig. 2. Entanglement swapping procedure using superconductive particles. Taken from [12].

III. QUANTUM INTRODUCTION IN CLASSICAL NETWORKS

Progress towards a quantum Internet will be gradual, and it starts by implementing quantum mechanisms and communications in modern-day infrastructure. In this section, we outline recent efforts in the experimental applications of quantum networks as well as common approaches for near-term hybrid quantum-classical networks. Finally, this thesis aims to define the obstacles quantumness brings to classical infrastructure.

A. Present and Near-Term Hybrid Implementations

As mentioned previously, quantum technologies (primarily QKD) are already beginning to be implemented experimentally using a combination of optical fiber and free-space optics (FSO). Indeed, networks have already been shown to be feasible across national and continental scales. In [13], at the time of publication, the authors claim the first integration of coherent quantum communication with the use of commercially available infrastructure. The link spanned over 250km, connecting two cities with the use of an untrusted quantum repeater and stateless design, meaning no use of

quantum memory, where physical qubits are held in storage until transmission or entanglement. In a similar vein, the authors of [14] accomplished a robust QKD network with the use of ground-satellite links totaling a distance of over 4,600 km.

In general, satellite-based networks are agreed to be the most optimal choice when introducing quantum capabilities in long-range communication, mainly due to reduced levels of noise when compared to terrestrial networks [3], [9], [15], [16]. One such conceptual framework places satellites at different atmospheric levels and, alongside optical fiber on the ground, uses line-of-sight connections via FSO links to establish connection between end users. Here, entanglement links would be dynamically established across each satellite layer. On the other hand, end-to-end entanglement is set up via swapping in response to user requests. Any extension to the range (say, across inter-continental distances) would be solved with the assignment of satellites as quantum repeaters and the inclusion of quantum memories [16].

An extension to this line of reasoning can be seen in the work provided by the authors of [3]. The proposal of the next generation in mobile communication network, dubbed 1Q, constructs an architecture based on the quantum counterparts of classical base stations and their operating regions. In fact, many parallels are drawn directly from services already provided in the classical domain. Quantum base stations (QBS) are visualized as taking the role of quantum repeater as well as providing quantum applications to users. Connectivity is modeled after the standard operation of 5G systems, where users, when powered up, are synchronized with the nearest classical base station (CBS) entering an idle state. Then, connection must be explicitly established with a QBS, where a joint physical and link layer effort generates and establishes entanglement between user and QBS.

Another common approach to a quantum network hybrid is with Software Defined Networks (SDN). This networking paradigm separates control logic from forwarding hardware, striving for increased flexibility through the simplification of network management [17], a commodity when working with quantum components. In a hybrid setting, this approach provides control over quantum links as well as the smooth integration of quantum (e.g. QKD) devices. In the architecture proposed in [15], SDN controllers manage both quantum and classical elements (including hardware and communication infrastructure) across data, control, and application planes, leading the way for potential hierarchical or distributed architectural designs. Moreover, by focusing on the pivotal resource that entanglement provides, this central component can be further specified as an entanglement-defined controller [18], opening the door for scalable quantum-native functioning.

B. Challenges

While these theoretical—and seemingly feasible—considerations have been given for the near-term integration of quantum technology using classical communication infrastructure, the complications that arise due to this interplay should not be understated. For that matter, this thesis aims to formalize the various issues that quantum mechanisms introduce to classical communication networks.

To begin with, the implementation of quantum networks is heavily dependent on the technologies that enable them. As mentioned above, quantum repeaters and memories are at the

forefront in unlocking reliable long-range transmissions. Additionally, given that different physical realizations of qubits are best suited for varying situations (e.g. photons are best used in transmissions, while trapped ions excel in storage implementations [7]), there is a critical need for quantum transducers. These devices convert qubits between their representations and are necessary for a complete realization of a quantum network. However, most of these technologies, given their novelty—and a lack of commercially available solutions—are not yet feasible for the wide-spread installations [19], [20], [21]. The only solution to this is the continued research and optimization of quantum technologies, as current quantum-empowered networks will largely be deployed with the minimum required components and a priority of function over form [15].

Regardless, hybrid networks' major selling point is the usage of existing infrastructure. Therefore, a point of concern exists in the ability for quantum communication to coexist alongside the transmission of classical information, i.e. in optical fiber. Specifically, quantum signals are inherently weak and, as a consequence of the no-cloning theorem, cannot be amplified on transmission. Classical signals, on the other hand, are considerably more powerful and present a risk of overshadowing quantum communications if they are in proximity, also referred to as "crosstalk." For this topic, common approaches include filtering techniques, various forms of multiplexing (wavelength, frequency, temporal), and resource allocation [19], [20], [21], [22], [23], [24]. In fact, these strategies have widely been successful, with coexistence being experimentally achieved within acceptable ranges.

However, a potential discord lies within classical components themselves which—although in a conventional network do not introduce significant noise—may compromise quantum communication. An example outlined in [19] is the optical amplifier. This component introduces noise through spontaneous emission, and, if sufficient care is not placed in its mitigation, it can be disruptive for a quantum network.

On the other hand, it should be noted that classical communications can be negatively impacted if hybrid networks are not carefully designed. Indeed, if certain networks are designed to prioritize quantum traffic (e.g. reserving link capacity) thereby treating classical links as an afterthought, the overall network capacity will suffer, as highlighted in [22].

Another concern lies in the precise synchronization necessary for quantum procedures. Decoherence is the predominant adversary in quantum communications as the qubit in its various forms is under constant threat of losing its quantum-ness (and with it, its utility) from a myriad of sources. While a portion of these come from the usage of a noisy channel—which is ultimately unavoidable—the other half come from synchronization demands held between quantum nodes, known also as quantum clock comparison. Its counterpart in classical communication is something relatively easily established using periodic reference signaling, but, due to quantum's intrinsic limitations, this strategy is not viable. Geographic distances are also relevant when facing this issue, as nodes physically distant from one another are at greater risk of missing this crucial mark [7]. Indeed, propagation becomes a factor to consider as the precision required for some quantum protocols is in the orders of femtoseconds, which is a figure unsatisfied by current classical network performance [9], [10]. Ultimately, whenever

these constraints are not met it leads to the irreversible failure of the process, which underscores the severity of this problem in the aim of a globalized network.

In the current state-of-the-art, strategies include using varying forms of multiplexed waveform pulses, photon temporal correlations, the White Rabbit system, and others to achieve necessary synchronization [7], [19], but it remains arguably quantum networking's most threatening hurdle.

Additionally, although quantum mechanics is heralded for its physics-backed security, a hybrid network is still vulnerable to malicious actors. This susceptibility arises due to imperfections in the devices used to enable security protocols like QKD, which can then be exploited by a potential eavesdropper in a field called quantum hacking [25]. In general, with imperfect quantum communication, sending and receiving parties monitor key parameters (e.g. quantum bit error rate) which are very consequential in QKD protocol. Given the inevitable errors that will occur in a given communication, communicating parties typically set a threshold security value that can alert them in case of an overly-insecure quantum channel. This window allows for an eavesdropper to intercept transmissions (thereby introducing additional loss into the channel) as long as the threshold is avoided.

Ultimately, countermeasures include systematic testing of one or more security parameters through tests that simulate hackers, similar to classical methods [25], [26].

While this final subsection has outlined several pressing points stacked seemingly against quantum's feasibility, it is important to note the potential that rests in continued research and development. Indeed, quantum information processing is likely on the precipice of rapid improvements and integration, as echoed in [3].

IV. THE DEVELOPMENT OF A DEDICATED QI PROTOCOL STACK

Regardless of the hurdles present in quantum technology, a quantum-enhanced internet is inevitable. Similar to the development (and eventual success) of abstractions like the OSI and TCP/IP models—and with ongoing research providing varying proposals for a quantum internet protocol stack—the goal of standardization is of utmost importance in the long term. Indeed, the use of these models allow users to implement intralayer solutions and alterations without the fear of impacting neighboring echelons in a principle called separation of concern. In this section, we seek to highlight the benefits that can be obtained from a quantum stack within the telecommunication field. Then, we survey proposals in literature on quantum network architecture and its principles, preceded by classical networking strategies that are reproduceable in the quantum domain. Concluding, this thesis aims to contribute to the higher-level discussion of design direction.

A. Quantum Benefits in the Classical Domain

First, it must be plainly stated that a quantum network cannot exist on its own. Indeed, simply by looking at the teleportation protocol outlined in section II, it is clear that no useful communication can take place without the classical system acting as a support. However, quantum is not the only beneficiary when both systems are integrated in concert;

classical communication stands a lot to gain from quantum networks, both directly through supporting quantum functionalities and indirectly as a byproduct of scientific advancements. We note that security enhancements are extensively documented throughout literature, so they will not be restated here.

Progressive efforts in the field of quantum information processing have significant overlaps in classical contexts. In fact, given that most short- and medium-term implementations aim to use existing optical fiber infrastructures, it is not unreasonable to conclude that technologies developed in the name of quantum advancement will find new applications in classical frameworks [19]. Particularly, research covering lower-loss communication components could have significant positive ramifications in the power, energy efficiency, and—arguably most importantly—cost considerations in classical communication links. Additionally, quantum’s incompatibility with most traditional technology may be enough motivation for a push towards all-optical classical networking, thereby receiving the benefits of said devoted research. As such, quantum research poses indirect benefits to classical infrastructure and may lead to a more efficient, and stronger, classical communication network.

Quantum resources also hold the potential to enhance several classical functionalities, particularly those found in the physical, data link, and network layers, as covered by [27]. Namely, the authors detail quantum’s ability to transmit classical information thereby surpassing limitations in channel capacity. Additionally, dealing with channel access, entanglement presents a collision-free strategy through exploitation of W states which, when measured, only one node observes the outcome $|1\rangle$ and is granted access to the communication resource. Finally, quantum-enhanced routing protocols could exploit superposition for simultaneous computation of an optimal end-to-end trajectory, allowing for improved transmissions and lower latency.

B. A Quantum Internet Protocol Stack

Given the uniqueness of quantum mechanisms and their limitations, most approaches in place for classical interconnectivity cannot be used in fielding its quantum counterpart. Consequently, a quantum protocol stack cannot be constructed one-to-one with current methodologies. On the other hand, the classical Internet stands as a valid point of reference in the design of quantum internet protocols. In certain cases, we can replicate classical paradigms and protocols while keeping quantum peculiarities in mind. This subsection covers such scenarios and highlights eminent protocol stack proposals as well as an uncertain topic regarding the optimal mode of entanglement in a network.

1) Replicable Classical Approaches in Quantum Systems

To start, the approach to networking is often visualized using graphs, i.e., a set of nodes and the connections between them, called edges. This abstraction holds when translating to quantum networking, and is particularly relevant when discussing how to most efficiently send a piece of information between two distant nodes, whereby the choice of an optimal path is called **routing**. This decision is taken through a routing algorithm, where we can easily borrow existing examples from the classical domain such as Dijkstra [28], which was conceived to find the shortest route in a

weighted graph. Indeed, these algorithms are dependent on the weights assigned to a graph’s edges, which, in quantum networking, is ultimately contingent on the application in use.

In literature, the choice in routing metric is a topic of debate [9], but is eventually linked to the quality of a particular entanglement (or qubit), also known as **fidelity**, which is marked by decoherence. Apart from this important distinction, Dijkstra’s algorithm can be used as-is [28]. Additionally, as early quantum systems will be characterized by their low quantum resources, it becomes prudent to implement another classical technique: traffic engineering. Indeed, this ensures routes are chosen carefully to guarantee any requested resource (i.e., the chosen routing metric) while keeping in mind its overall availability in a system.

Furthermore, oversimplifying, once we arrive to the actual transmission of a given signal after a routing decision, another useful classical technique includes **multiple input multiple output (MIMO)** techniques. These, credited with revolutionizing classical wireless systems, are able to improve reception quality or increase attained throughput when applied to quantum communication systems—an experimental claim highlighted in [20]. Particularly, one such technique is through spatial filtering, also known as beamforming, which is a way to angle signals towards (or away from) a sensor, e.g., an antenna. This practice improves the signal-to-noise ratio of an optical signal—typically in the terahertz range for QKD—and can lead to array gain through constructive wave interference. As a result, this can lead to increased transmission distance and greater access to quantum applications.

Then, given our main concern with quantum transmission is of error in the form of decoherence, several strategies developed for classical systems can be adapted nearly directly to quantum communications. First, **quantum error correction coding (QECC)** has advanced thanks to parallels with classical techniques which adds redundancy to transmitted information qubits. This redundancy, in the form of added qubits, allows the receiver to exploit the structure of the code and correct any potential errors in transmission, exactly like classical error correcting code. In fact, the breakthrough work (as it was initially considered impossible in the quantum domain) presented a 9-qubit QECC able to correct a single qubit. However, this added complexity is difficult to implement in current state-of-the-art due to scarcity of usable qubits [20].

On the other hand, a surprisingly reproduceable classical protocol is ARQ, which is a form of error mitigation. Through the use of the Universal Quantum Copying Machine (UQCM), transmissions can be bolstered through the creation of imperfect copies of a given quantum state. Like its classical counterpart, quantum ARQ uses acknowledgement and timeout messages to establish semi-reliable communication through retransmissions on losses [29]. Admittedly, given the lower quality of the “copied” qubits, this strategy is only viable for applications who do not require high fidelity qubits. In a similar vein, the development of a quantum TCP (qTCP) resulted in a three-stage connection (handshake) protocol aimed at preventing the loss of quantum packets, similar to its classical counterpart [30]. This protocol splits a given quantum message into three fragments such that, if two are lost, the remaining fragment can divide into three new shares for retransmission.

Finally, while aiming to manage multiple users in a future quantum internet, medium access control (MAC) protocols are critical to define as they also mitigate losses in communication. Research in this area is sparse, but classical techniques such as slotted ALOHA, which is based on designated time slots for transmission, and code division multiple access (CDMA), which assigns codes to transmitters without temporal constrictions, have been outlined [31], [32]. The former can be directly translated to quantum communications, where each contending node holds an entangled W state qubit. Upon measurement at the start of a time slot, only one node will observe a $|1\rangle$ state and be granted access. Indeed, most temporal-based classical MAC protocols have straightforward translations in quantum networking systems due to their relative simplicity and, ironically, lack of dependence on quantum resources.

Furthermore, it is important to remember how quantum teleportation—and thereby a significant portion of quantum communication—is based on entanglement where measuring one entangled qubit has an immediate effect on all other related qubits. As such, the idea of “collision” strictly in entanglement-based communication holds little meaning as information can only flow in one direction at a time. Regardless, particularly when focused on non-entanglement quantum transmission, techniques such as a quantum CDMA protocol (which, in [31], barcodes photons) are relevant in enabling other quantum applications. Here, a quantum signal can be represented in its corresponding frequency domain and multiplied with an assigned encoding operator. Upon reception, the encoded signal can be reassembled through inverse operations (e.g., Fourier transformations).

2) State-of-the-Art Overview of Proposals

While useful, individual protocols only make up a part of a complete stack. In the current state-of-the-art, there exist several proposals for a dedicated quantum internet protocol stack. Particularly, we will briefly state a tenet in the creation of a quantum Internet. Then, we highlight three leading architectures which are highly referenced in literature, as well as one additional, promising proposal.

One of the first comprehensive approaches to the design of a quantum internet protocol stack is by Van Meter et al. [34]. It centers itself on quantum repeaters and a bipartite entanglement structure (i.e., the usage of EPR pairs between nodes). The proposed architecture is composed of five layers, and closely mimics the structure of the OSI model. Indeed, the authors detail a physical layer, tasked with the generation of EPR pairs between adjacent nodes, and a link layer (called the entanglement control layer) which keeps track of the success or failure of said single-hop entanglement. Then, upper layers, composed of purification control (PC) and entanglement swapping control (ESC), are responsible for end-to-end level connection as they are repeated in tandem for each power of two distance. Specifically, PC is in charge of specifying where purification is needed for maintaining high fidelity, and ESC is tasked with swapping, alongside success/failure communications to end nodes. This culminates in the application layer, which would include any application of quantum mechanisms with high fidelity pairs guaranteed by the architecture.

Ultimately, after incremental extensions and refinements, [35] details a recursive network paradigm (dubbed Quantum Recursive Network Architecture) alongside a bootstrap

protocol for improved fidelity accountability. This work accomplishes goals of scalability as well as inter-networking and promises support for multipartite entanglement and the hardware capabilities of the present and future—one of the aforementioned principles of a quantum internet. In fact, although being one of the first proposals, Van Meter’s model is still solid in its contributions [10].

Next, Wehner et al. [36], [37] present a protocol stack with a focus on decoherence mitigation based on bipartite entanglement, aiming to provide a building block for more complex quantum network services. The proposed architecture contains five layers with names directly matching the joint OSI-IP/TCP model: physical, link, network, transport, and application. The authors detail a quantum data plane protocol responsible for end-to-end entanglement, where robust entanglement generation is tasked to the physical and link layers while the network layer is in charge of entanglement swapping for endpoint communication as well resource management. Notably, the authors detach the bottom two layers (which are heavily reliant on hardware specifications) from the rest of the protocol stack, creating an abstraction then leveraged to ensure the proposed model is hardware-independent—an important property for long-term flexibility and scalability. Additionally, through the use of a cutoff time which discards decohered qubits, experimental validation highlights the establishment of successful communication at requested fidelity rates, detailed in [37].

Extending upon this, in [38], the authors utilize the above architecture as a foundation for setting up high-level software in the implementation of a quantum network operating system—named QNodeOS—which then led to the development of an application environment called Qoala (which runs on QNodeOS) [39]. Most recently, [40] tackles the issue of scheduling in such an execution environment by introducing a structure which determines when entanglement generation can take place. This provides a unified approach to the timings of quantum applications at end nodes as well as coordinating the use of resources within a single quantum network.

Third, continuing the list of state-of-the-art architectural proposals, Dür et al. [41] distinguish themselves from the previous two proposals through the adoption of multipartite states instead of EPR pairs, as well as detailing a four-layered architecture comprised of physical, connectivity, link, and network layers. Here, the physical layer is in charge of both generation and transmission of quantum resources through physical channels, as well as short-link entanglement between nodes at the lowest level. Instead, the connectivity layer oversees imperfections generated in the previous layer as well as the long-distance entanglement of end users. Then, the link layer guarantees a constantly-refreshed entangled network state for dynamic handling of incoming requests. Finally, the network layer is responsible for connections with external networks through the quantum counterpart of a classical router.

However, a recent work co-authored by the main research pair of [41] revisits the approach detailed in above, proposing to move away from OSI-style stacks and instead aim for resource-centric, task-based schemes [42]. In such a structure, a request is first received from a quantum application for a particular service (e.g., entanglement

generation or qubit transmission). Then, the quantum network, through the use of a centralized resource manager, derives a distributed workflow, referred to as saga. While interesting, this proposal is largely self-contained as it does not detail how it would fit in today's (or future) telecommunication infrastructure, but it may serve as a beginning of more research on non-OSI inspired models.

Finally, Fan et al. [43] present a protocol stack which, to their knowledge and ours, presents the first dedicated cross-layer coordination model. The work separates the planning and execution of multipartite entanglement-based communication through the introduction of a Global Entanglement Module (GEM). This module is in charge of synchronizing and providing nodes across the network with near-real-time entanglement availabilities, thereby supporting a decentralized model for adaptive, context-aware decisions. As such, this proposal presents a six-layered design as seen in Fig. 3. Particularly, this proposal specifies the use of pre-distributed entanglement (planned in the transport layer, executed in the swapping layer) in an effort to reduce latency during execution. Indeed, since the first two layers do not introduce any new features, the bulk of features are found between the distributed entanglements layer and the swapping/fusion layer. The former serves as the planning core of the stack and specifies how entanglement should be established, providing instructions to the swapping layer which executes—and adapts, according to information provided by GEM—in real-time while focusing on fidelity and latency.

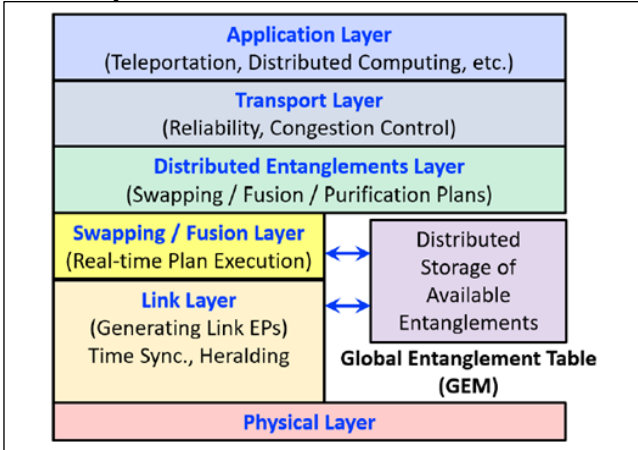


Fig. 3. The six protocol layers as proposed by Fan et al., taken from [43].

The comparisons between the first three models have been extensively covered in existing literature [10], and interested readers should review [21] for a detailed layer-by-layer breakdown.

3) Consensuses and Points of Contention on Design Philosophies

Concluding, this thesis will cover a contentious topic in the design philosophy of quantum internet networks.

Foremost, it is important to stress that the most important resource in quantum networks is entangled groups. Indeed, as mentioned previously, long-range quantum communications are impractical without it, and, crucially, their states are known meaning the operation can be retried without any safety concern, as entangled bits on their own do not leak any information. As a result, one of the most consequential

decisions when designing a quantum internet architecture is based on when entanglement should be established. This topic is widely debated, with researchers split between on-demand and pre-distributed entanglement, also known as reactive and proactive distributions, respectively.

The former, as the name suggests, establishes entanglement between nodes upon request. Given that current and near-future quantum systems will run on low quantity and quickly decohering resources, it is very likely that this format will be established in an attempt to guarantee service to all users. However, this mode comes with its disadvantages. Most notably, the main downside of reactive distribution is the resulting latency of a given connection. Indeed, entanglement is ultimately an unreliable service, for both swapping and purification are not guaranteed successes given they are probabilistic in nature, even under ideal circumstances. On the other hand, on-demand entanglement means routing paths can be calculated before the generation and distribution of quantum resources, thereby preserving said resource [9].

Thus, once quantum systems mature (particularly quantum memories), it may make more sense to establish proactive entanglement systems. Here, each possible virtual link between nodes is constantly re-established to ensure a complete graph state. This results in quicker communications, as the main ingredient for teleportation has already been distributed. Additionally, with such a dynamic topological state as entangled pairs decohere and are regenerated, routing decisions are based on existing links. Pre-distributed systems integrate recent topological data, but are at risk of the quality of links established. This, however (as foreseen in [43]), can be mitigated through higher level controllers. Furthermore, by allowing for a near-constant state of connection in a network, it leads to the concepts of a quantum address and quantum paths (a detailed review can be found in [18]).

Regardless, as noted in [3], systems can be designed to fit both approaches, depending on what is best for the network and its users at a given time. Indeed, a network (with potential help from predictive technologies) could determine which links can stay disconnected in an effort to conserve energy and minimize resource usage. Otherwise, if there was a need for reduced latency and increased quality of service, networks could transition to a more proactive approach. This leads to a deeper delve into the topic of scheduling, which can be further read in [9], [44] which are inherently dependent on the demands placed on quantum resources.

V. CONCLUSION AND FUTURE DIRECTIONS

Quantum technology and the idea of a future quantum Internet are both in their infancy. With recent resurgence in research, we advance towards the goal of global access to quantum applications and their unparalleled services, some of which are in place today. This thesis provided a briefing on the current state-of-the-art developments for quantum internet networks. By reviewing literature, and avoiding mathematical intricacies, we hope to support research in both physical implementations and higher-level network abstractions by presenting the current state of study.

However, as a still-emergent field, there is a lot of work to be done. Particularly, quantum enabling technologies, like quantum repeaters, quantum memories, and the physical

realizations of qubits, must take form. Additionally, as showcased in this thesis, there are multiple proposals with varying ways of handling quantum interconnectivity, yet no clear direction on what are the most optimal design choices. Therefore, additional research must be devoted to facilitating comparisons between models, as well as the determination on the most adequate metric that allows for a fair quantitative comparison.

Finally, a larger point of discussion is whether a layered perspective is the correct choice in quantum networking architecture. This line of thinking stems from the increasingly obvious need for flexibility and cross-layer interactions which are at odds with conventional stack guidelines as layers can only communicate with those adjacent to it. Additionally, given the ethereal nature of quantum resources, having structures with significant protocol overhead could be detrimental in the quick processing of quantum information. However, it is undeniable that some form of standardization is necessary for global scalability. Efforts for international standardization are underway, and global collaboration is critical for the sophistication of future research and, eventually, a quantum connected society.

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